Testing a Proposed Rotational Pole Against Earth's Preferred-Axis Geometry

A preferred-axis constraint screen

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Abstract

This case study tests whether the proposed ECDO pole $N_p' = (31°\mathrm{E}, 14°\mathrm{S})$ is supported by preferred axis patterns in mantle tomography, static gravity, the solid lithosphere, and GRACE/GRACE-FO degree 2 mass variability. The lowermost mantle axes form a statistically significant cluster in three separate null tests with 10,000 realizations each. The family balanced center is located near $1.02°\mathrm{S}, 17.24°\mathrm{E}$, placing N_p' 18.81° away. This distance is well outside the empirical 99% family bootstrap radius of 3.04°. Some individual results lie closer to N_p', including the primary static gravity maximum axis at 9.45° and one UNICA25 lowermost mantle tensor at 2.16°. These closer alignments are reported as secondary descriptive results. The main pooled lowermost mantle principal axis test does not recover N_p' within its family bootstrap region. These results describe geometry only. They do not establish mantle displacement, physical coupling, dynamics, causation, or ECDO.

Keywords: inertia tensor; principal axis; mantle tomography; static gravity; GRACE; axial statistics; candidate-pole testing

1 Introduction

If Earth's mass distribution has a persistent preferred direction, some trace of that direction should appear across related systems: the deep mantle, the present-day gravity field, the solid lithosphere, and the time variable degree 2 gravity field measured by GRACE and GRACE-FO. These systems sample different layers and timescales, so their axes are not expected to match exactly. They can still be compared using the same geometry, sign convention, and decision rule.

This analysis is especially relevant to the proposed Exothermic Core-Mantle Decoupling-Dzhanibekov Oscillation, or ECDO, framework developed by Roger B. Cunningham [1]. ECDO proposes that Earth's mantle and crust could undergo a large reorientation relative to the rotational axis under specific core-mantle coupling conditions. Cunningham describes the broader theory in *Inversion*, while The Ethical Skeptic outlines a modeling approach that treats the proposal as a constrained physical system whose geometry, energy, torque, and coupling requirements can be tested separately [2].

The ECDO geometry examined in this project includes a proposed mantle and crust displacement of approximately 104° and a proposed new-pole location at N_p' [1]. Before considering how such a displacement could occur, it is useful to ask whether Earth's present gravity and inertia geometry points toward that location.

This report evaluates one part of that larger problem: *gravity and inertia geometry.* It does not test the proposed trigger, the coupling between Earth layers, the displacement dynamics, or whether an ECDO event occurred in the past. Those questions require separate investigations.

Gravity and inertia geometry is also only one evidence class in the broader candidate-pole framework. Other classes include paleomagnetic and virtual geomagnetic pole data, plate kinematics, hotspot reference frames, paleoclimate, sea level, tectonics, volcanism, biogeography, archaeological bearings, and mechanism feasibility. Each class should be evaluated on its own terms. The purpose here is to determine what the combined gravity and inertia results support, which alignments remain secondary, and how the proposed coordinate compares with the main recovered geometry.

The candidate tested in this report is the project-local coordinate N_p', fixed at

$$N_p' = (31°\mathrm{E}, 14°\mathrm{S}).$$

The test is:

> Does N_p' fall within the preferred-axis region recovered from the gravity and inertia data when every comparison uses the same declared statistical and physical conventions?

2 Model setup

2.1 Evidence components

The analysis combines four independent sources of preferred-axis geometry:

1. **Mantle tomography:** principal-axis tensors derived from eight global mantle models, evaluated over five depth intervals together with lower-mantle and LLSVP-like summaries.

2. **Static gravity:** degree-2 trace-free tensors computed from the declared GOCO2025s static gravity residuals using the reference-compatible residual convention [15, 16].
3. **Solid lithosphere:** the LITHO1.0 solid-lithosphere degree-2 tensor and its three principal axes [4].
4. **Time-variable gravity:** monthly degree-2 GRACE and GRACE-FO tensors together with principal-axis summaries, unordered eigenframe comparisons, and branch-tracking tests [17, 18, 19, 20, 21].

The mantle comparison includes five widely used seismic-velocity tomography models: S40RTS [5], TX2011 [6, 7], SEISGLOB2 [8], SPani [9, 10], and SEMUCB-WM1 [11]. Two fields from the Collaborative Seismic Earth Model (CSEM2-Globe), the shear-wave velocity model (VSV) and the density-related field (RHO), are also included [12, 13], together with the UNICA25 absolute-V_p model [14].

Each mantle model is evaluated over five depth intervals (0–410 km, 410–660 km, 660–1500 km, 1500–2500 km, and 2500–2891 km). Whole-mantle and lower-mantle summaries are also reported because they combine information from multiple depth intervals into a single preferred-axis comparison.

2.2 Trace-free inertia geometry

For an anomaly field with effective density perturbation $\delta\rho(\mathbf{r})$, the anomaly inertia tensor is

$$\Delta\mathbf{I} = \int_V \delta\rho(\mathbf{r}) \left(r^2\mathbf{1} - \mathbf{r}\mathbf{r}^{\mathsf{T}}\right) \, dV. \tag{1}$$

Only the trace free degree 2 part is used to determine the principal axes. Here, *trace free* means that the equal, direction independent part of the tensor has been removed, leaving only the directional differences that control the axis orientations:

$$\mathbf{Q} = \Delta\mathbf{I} - \frac{\mathrm{tr}(\Delta\mathbf{I})}{3}\mathbf{1}. \tag{2}$$

The symmetric tensor is diagonalized as

$$\mathbf{Q}\mathbf{e}_i = \lambda_i\mathbf{e}_i, \qquad \lambda_{\min} \leq \lambda_{\mathrm{mid}} \leq \lambda_{\max}. \tag{3}$$

Each eigendirection is axial, so $\mathbf{e}$ and $-\mathbf{e}$ represent the same axis. The axial angular distance between two directions is therefore

$$d_{\mathrm{axial}}(\mathbf{a}, \mathbf{b}) = \cos^{-1}\left(\left|\hat{\mathbf{a}} \cdot \hat{\mathbf{b}}\right|\right). \tag{4}$$

On a map, the two ends of the same axis can appear far apart even though they represent the same axial direction. The final candidate pole map therefore shows the end of each axis that is closest to the candidate. This affects only how the axes are displayed and does not change the calculated distances.

2.3 Physical role versus unordered geometry

The report uses two different kinds of axis comparison:

Preferred rotational axis. The principal axis with the largest moment of inertia under the adopted physical sign.
Unordered principal direction. A geometry only comparison in which the minimum, intermediate, and maximum axes may exchange roles.

The unordered comparison tests whether two tensors have a similar overall principal axis pattern, even when the axis roles are different. It is not treated as the preferred rotational axis.

A candidate may lie close to one axis in the unordered frame while remaining far from the physical maximum-moment axis. For this reason, the final interpretation keeps axis geometry and axis role separate.

2.4 Static-gravity declaration

The primary static-gravity residual is frozen as

`static::R_compatible_central`,

described in the project outputs as the *reference-compatible central partial residual.* Its maximum-moment axis is the primary static-gravity comparison.

Two additional residuals are retained as sensitivity cases:

`static::R0_reference_residual`,
`static::R_anomaly_sensitivity`.

These residuals answer related but non-identical reference-state questions. The primary result is not allowed to move between them after seeing which one lies closest to the candidate.

2.5 Tomography-to-density sign

Tomography gives the directions of the principal axes, but the density sign determines which direction is labeled minimum or maximum moment. The primary interpretation used here follows the gravity-supported sign test reported in [24]. In that study, seven of the eight full-mantle model fields favored the globally reversed sign in the primary degree-2 gravity comparison.

The opposite sign is also kept as a sensitivity test. Reversing the sign gives

$$\mathbf{Q} \to -\mathbf{Q}. \tag{5}$$

This does not move the principal axes. It only changes their roles: the minimum axis becomes the maximum axis, the maximum becomes the minimum, and the intermediate axis stays intermediate. The overall axis geometry therefore remains the same even though its physical interpretation changes.

2.6 Family balancing and pooled center

The lowermost mantle group includes the 2500-2891 km tensor and the LLSVP-like subset when both are available. Some model families contain more models or tensors than others, so equal weighting would allow those families to have more influence. To avoid that, the weights are divided in three steps:

$$w_{ijk} = \frac{1}{N_{\text{family}}} \frac{1}{N_{\text{model|family}}} \frac{1}{N_{\text{tensor|model}}}. \tag{6}$$

Each family receives equal total weight. Within each family, the weight is divided equally among its models. Within each model, the weight is divided equally among its eligible tensors.

For a trial axial center $\boldsymbol{\mu}$, each tensor contributes whichever of its three principal axes lies closest to that center. The selected axes are combined in the weighted orientation matrix

$$\mathbf{A} = \frac{\sum_j w_j \mathbf{u}_j \mathbf{u}_j^{\mathsf{T}}}{\sum_j w_j}. \tag{7}$$

The leading eigenvector of $\mathbf{A}$ gives the family balanced pooled center. The final lowermost-mantle set contains 15 eligible tensors from eight models in four model families.

2.7 Null models

Three null tests were run with 10,000 realizations each:
Depth permutation. Shuffles the depth labels within each model while keeping the same model and family structure.
Independent model rotation. Randomly rotates each model in three dimensions while preserving the spacing between those three principal axes.
Shallower-interval substitution. Replaces the two lowermost-mantle tensors with two different shallower-depth tensors from the same model when available.

The observed clustering is measured in three related ways:

$$C_{\text{axial}} = \lambda_{\max}(\mathbf{A}), \tag{8}$$

$$D_{\max} = \max_{i<j} d_{\text{axial}}(\mathbf{u}_i, \mathbf{u}_j), \tag{9}$$

$$\overline{D}_P = \frac{2}{n(n-1)} \sum_{i<j} \frac{\|\mathbf{u}_i \mathbf{u}_i^{\mathsf{T}} - \mathbf{u}_j \mathbf{u}_j^{\mathsf{T}}\|_F}{\sqrt{2}}. \tag{10}$$

The axial concentration, C_{axial}, increases when the selected axes cluster more tightly. The maximum separation, $D_{\max}$, is the largest axial distance between any two selected axes. The mean projector distance, $\overline{D}_P$, measures the average difference between all pairs of axes while treating opposite ends of the same axis as equivalent.

Stronger clustering gives a larger C_{axial} and smaller values of $D_{\max}$ and $\overline{D}_P$.

Empirical p-values are calculated as

$$p_{\text{emp}} = \frac{k+1}{N+1}, \tag{11}$$

where k is the number of null realizations at least as extreme as the observed result and N is the total number of realizations. This correction prevents a reported p value of exactly zero.

2.8 Bootstrap containment rule

Uncertainty in the lowermost mantle center is estimated with 2,000 bootstrap samples. In each sample, one model is selected from each model family. The candidate is then classified by its axial distance from the final family balanced center:
Strongly consistent: inside the empirical 68% radius.
Consistent: outside the 68% radius but inside the 95% radius.
Tension: outside the 95% radius but inside the 99% radius.
Unsupported: outside the 99% radius.
Indeterminate: the required data or provenance are missing.

In this run, the 95% and 99% radii are the same because the bootstrap results are discrete. The 2,000 samples produced only eight unique center locations and eight unique distances from the final center. The matching 95% and 99% values therefore do not represent two separately resolved uncertainty limits.

2.9 GRACE and GRACE-FO

Monthly degree-2 gravity changes are taken from the GFZ GRACE RL06 and GRACE-FO RL06.3 Level-2 GSM products, with the archived GFZ RL06 satellite-laser-ranging replacement series for $\bar{C}_{20}$ [19, 20, 21]. The monthly coefficients are converted into trace-free tensors and separated into their three principal axes. The results are summarized in three ways:

1. the named minimum, intermediate, and maximum axes;
2. an unordered comparison that allows the axis roles to switch;
3. branches tracked from one month to the next.

Uncertainty was tested with 1,000 block-bootstrap samples using six month blocks. Only diagonal coefficient uncertainties were available, so full covariance between coefficients could not be included.

The same axis branches could not be matched reliably between GRACE and GRACE-FO using 6-, 12-, or 24-month boundary windows. For that reason, the tracked branches are not treated as evidence that one physical axis continued unchanged across both missions.

2.10 Frozen candidate registry and provenance

The candidate registry was fixed before the final comparison. It contains one included target, shown in Table 1. The coordinates and source information were recorded in the frozen publication inputs so the exact candidate definition can be verified.

Table 1: Frozen candidate-pole registry used in the final comparison.

ID	Candidate	Latitude	Longitude	Source	Provenance and status
`np_prime`	N_p'	$-14.0°$	$31.0°$	Project-local target definition	Defined after the tensor calculations and axis-matching steps were completed; geographic latitude and longitude in degrees east; included in the final comparison.

3 Main results

3.1 The lowermost cluster is non-random

The lowermost geometry is tightly clustered across the final null tests. The observed statistics are

$$
\begin{aligned}
C_{\mathrm{axial}} &= 0.94619, \\
D_{\mathrm{max}} &= 42.98°, \\
\overline{D}_P &= 0.24511.
\end{aligned}
$$

Table 2 reports the final empirical probabilities. The most conservative value among the nine related diagnostics is 0.0106. These are converging measurements of the same clustering behavior, not nine independent discoveries.

Figure 1 shows the results of the three null tests. The colored distributions show the results expected under each null model, and the dashed black lines show the observed lowermost-mantle result. The observed axial concentration is larger than almost all null results, while the observed maximum separation and mean projector distance are smaller. All three measures show that the lowermost-mantle axes are more tightly clustered than expected under the tested null conditions.

This is the strongest result in the entire analysis. The lowermost set contains a shared axial tendency that is difficult to reproduce by relabeling depths, rotating each model independently, or substituting shallower mantle intervals.

The null result establishes unusual geometry under the declared models. It does not establish physical coupling, a migration path, a reorientation mechanism, or causation between layers.

3.2 Family-balanced center and bootstrap region

The calculated family balanced center is

$$(1.0207°\mathrm{N}, 197.2388°\mathrm{E}).$$

Because an axis has two opposite ends that represent the same direction, the equivalent end closest to the candidate is

$$(1.0207°\mathrm{S}, 17.2388°\mathrm{E}).$$

Table 3 summarizes the bootstrap results.

Figure 2 shows the 2,000 family-bootstrap centers and the location of N_p'. The bootstrap centers form a very small cluster around the final family-balanced center. Many points overlap because the bootstrap produced only eight unique center locations. The candidate lies well outside the 68%, 95%, and 99% regions.

The 95% and 99% radii are both 3.04° because the bootstrap distribution has only eight unique distances. These matching values do not represent two separately resolved uncertainty boundaries.

The center changes only slightly when different models are selected from the existing model families. This result depends on the chosen family groups and the limited number of available models. The bootstrap does not include the full uncertainty in the tomography models, mineral physics, depth-dependent density conversion, or covariance between models.

3.3 Direct candidate comparison

Table 4 shows the main candidate test and the closer secondary alignments separately.

Figure 3 shows the candidate and the main axis results on one map. It includes the family balanced lowermost center, the bootstrap centers, the static gravity maximum axes, the model family centers, the depth interval centers, the individual lowermost tomography axes, and the three solid lithosphere axes.

Each axial direction has two equivalent ends. The figure shows whichever end lies closest to N_p'. This plotting choice makes the results easier to compare and does not change any calculated distance. The figure shows that several individual axes lie near the candidate, while the family-balanced lowermost center remains farther away.

The main result is that N_p' lies well outside the family balanced lowermost mantle bootstrap region. The UNICA25 tensor at 2.16° and the primary static gravity maximum axis at 9.45° are still reported because they show that some individual results lie near the candidate. These are secondary alignments within a pooled geometry whose center remains farther away.

3.4 Global distance from the lowermost center

Figure 4 shows the axial distance from every point on Earth to the family-balanced lowermost center. Yellow areas are closest to the center, while darker areas are farther away. The maximum possible axial distance is

Table 2: Final lowermost-mantle null-test results. Each null used 10,000 realizations.

Null construction	$p(C_{\text{axial}})$	$p(D_{\max})$	$p(\overline{D}_P)$
Depth permutation	0.01060	0.00640	0.00100
Independent model rotation	0.000200	0.000100	0.000100
Shallower-interval substitution	0.000300	0.00280	0.000100

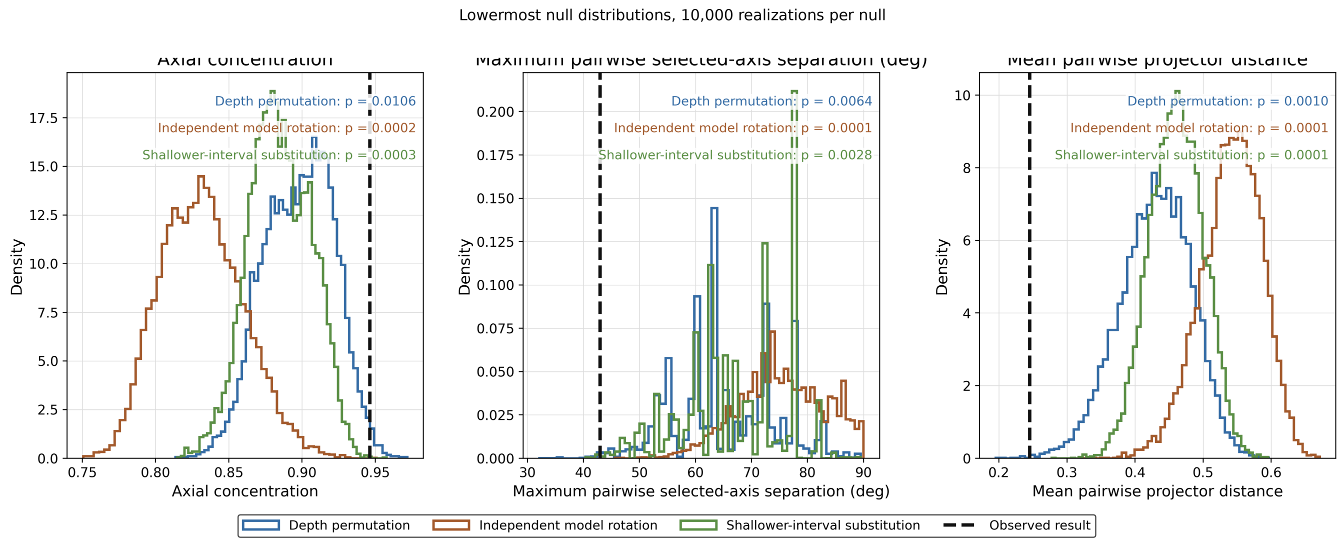

Figure 1: The observed lowermost-mantle clustering compared with three null tests. Each null test used 10,000 realizations. The dashed black lines show the observed results.

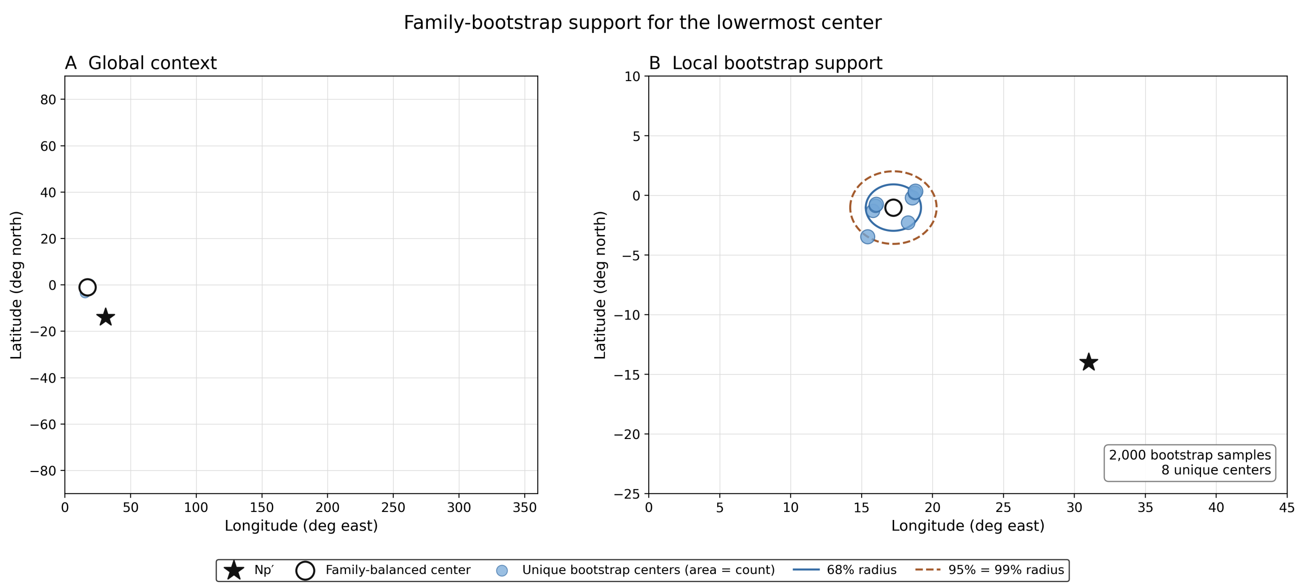

Figure 2: The 2,000 family-bootstrap centers, the final pooled center, and N_p'. Only eight unique center locations occur, so many bootstrap points overlap.

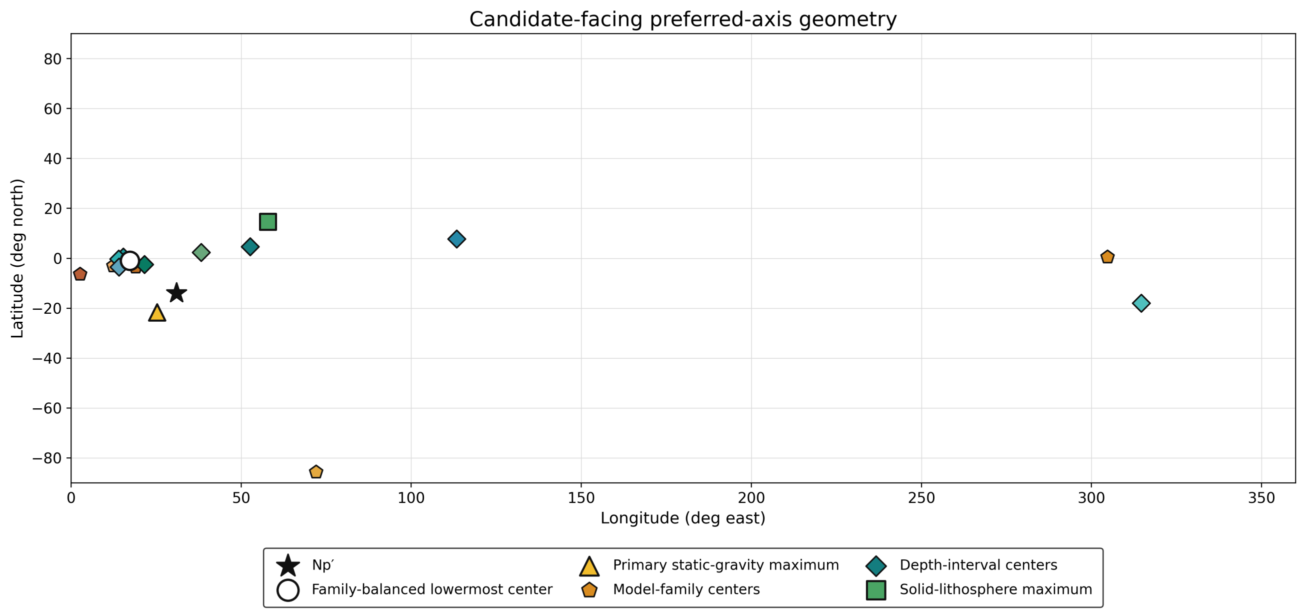

Figure 3: The candidate and the main gravity-and-inertia axis results. Each axis is shown using the end closest to N_p'.

Table 3: Stability of the family-balanced lowermost-mantle center.

Quantity	Result
Bootstrap realizations	2,000
Unique bootstrap centers	8
Median displacement	1.64°
68% radius	1.95°
95% radius	3.04°
99% radius	3.04°
N_p' distance	18.81°
Containment class	Unsupported

90° because opposite ends of an axis represent the same direction.

The white contours mark equal axial distances from the lowermost center. The contours wrap across the 0° and 360° longitude boundary because these longitudes represent the same location on the map. A second low-distance region appears near the antipode because the center and its opposite end represent the same axis.

The symbols show N_p', the primary static-gravity maximum axis, the model-family centers, and the depth-interval centers. The candidate lies 18.81° from the lowermost center. Several secondary results lie in the same broad geographic region, although they do not fall inside the small family-bootstrap region shown in Figure 2. The white contours in Figure 4 show geometric distance only; they are not confidence or uncertainty regions.

3.5 Static gravity residual sensitivity

The static-gravity result changes depending on how the residual field is defined. For the primary reference-compatible central residual, the maximum axis lies 9.45° from N_p'. The anomaly-sensitivity residual points toward the same broad longitude range, while the R0/reference residual lies much farther from the candidate.

This means the gravity comparison depends on how the reference figure and shallow mass contributions are removed. The primary residual was chosen before the final comparison so the analysis would not simply select whichever residual happened to lie closest to N_p'.

3.6 GRACE/GRACE-FO geometry

The GRACE and GRACE-FO mission averages have similar overall axis geometry, but the minimum and maximum roles switch between the two missions. The intermediate axis remains close. Table 6 summarizes the bootstrap results.

The mission averages are calculated relative to the same full-series mean. Because there are only two mission periods, this centering can make their average anomaly tensors point in opposite signs. The axis directions can then stay the same while the minimum and maximum labels switch. For that reason, the close mission-average geometry is not treated as separate proof that one physical axis persisted through time.

The month-to-month branch matching is also sensitive to the chosen boundary window. The 6-, 12-, and 24-month windows give different branch assignments and large angles across the mission gap. The supported conclusion is that the two missions have similar unordered axis geometry. A continuous physical branch across the gap has not been established.

Table 4: Direct gravity-and-inertia comparison for the frozen candidate N_p'. The lowermost-mantle bootstrap test provides the main classification. The other distances are secondary geometric comparisons.

Comparison	Axial distance	Role or source	Interpretation
Family-balanced lowermost center	18.81°	Selected-axis center	Outside the empirical 68%, 95%, and 99% bootstrap regions. The primary classification is unsupported.
Primary static-gravity residual	9.45°	Maximum-moment axis	Closest axis from the primary static-gravity residual. This distance is reported separately because the lowermost-mantle bootstrap does not measure gravity-field uncertainty.
Closest individual tomography tensor	2.16°	UNICA25 absolute V_p, 2500-2891 km	A close result from one model and depth interval. It does not replace the pooled family-balanced classification.
Closest pooled depth-interval center	14.85°	Depth-center comparison	Lies in the same broad region as the candidate.
Closest model-family center	15.82°	Family-center comparison	Lies in the same broad region as the candidate.
Solid-lithosphere maximum axis	39.09°	Preferred rotational axis	The solid-lithosphere maximum axis does not recover the candidate.
GRACE/GRACE-FO	Descriptive	Named and unordered frames	The same branch could not be tracked reliably across both missions, so no persistent physical axis is claimed.

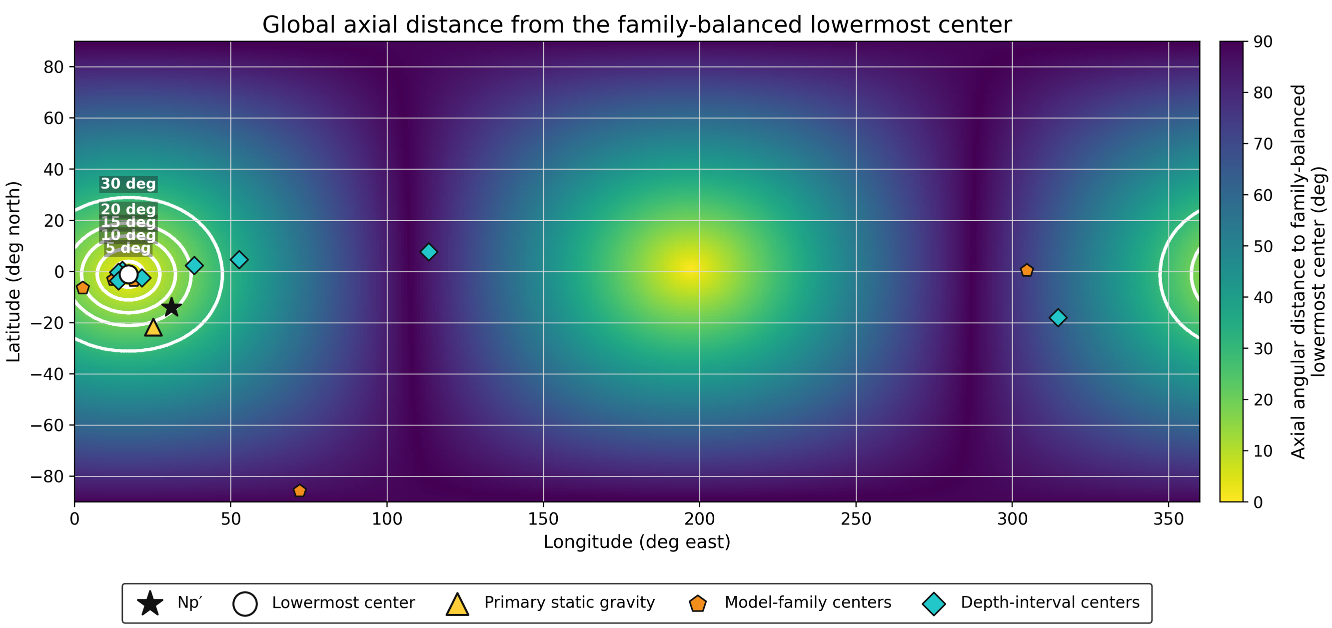

Figure 4: Axial distance from every location on Earth to the family-balanced lowermost center. Yellow areas are closest to the axis, and darker areas are farther away. White contours show equal-distance lines.

Table 5: Static-gravity residuals used in the analysis.

Residual	Analysis role
`R_compatible_central`	Primary reference-compatible partial residual.
`R_anomaly_sensitivity`	Sensitivity test.
`R0_reference_residual`	Reference-state sensitivity test.

3.7 Cross-validation and hierarchy claims

The cross-validation tests compared three possibilities: one shared geometry, separate geometries for each component, and several distinct branches. The multi-branch model performed slightly better when individual tomography models were left out. The single shared geometry performed better when entire model families or physical components were left out. This means the multi-branch pattern does not predict new groups consistently.

The results support a broad shared geometry, with some branch differences within the tomography models. They do not support a stable four-layer hierarchy connecting the lowermost mantle, static gravity, GRACE, and the solid lithosphere.

3.8 Distance from N_p'

The main result is:

> **N_p' is not supported by the family-balanced lowermost-mantle containment test.**

The candidate is 18.81° from the family-balanced center. This is well outside the empirical 99% bootstrap radius of 3.04°, so the final classification is `unsupported`.

Two individual results are closer:

> The candidate is 9.45° from the primary static-gravity maximum axis and 2.16° from one UNICA25 lowermost-mantle tensor.

These closer alignments are reported separately. They do not change the main classification.

4 Sensitivity and constraints

4.1 What changes the interpretation

Axis role. The result depends strongly on which principal axis is being compared. The physical maximum-moment comparisons do not overturn the unsupported classification from the pooled lowermost selected-axis test. In an unordered comparison, another principal axis may lie much closer. The unordered test compares overall axis geometry, while the physical test compares the maximum-moment rotational axis.

Static-gravity residual. The static-gravity axis changes when the reference figure or shallow mass corrections change. The 9.45° distance applies only to the declared primary residual and should always be reported with that condition.

Model family and branch selection. Some global and anisotropic S-wave model families select a low-latitude branch near the pooled center. Other model families select different, but still internally consistent, branches. Family balancing gives each model family equal influence, so families with more models do not control the result. The bootstrap still depends on how the available models are grouped into families.

GRACE branch identity. GRACE and GRACE-FO have similar unordered axis geometry, but the minimum and maximum roles switch between missions. The branch matching also changes when different boundary windows are used. A continuous physical branch across both missions has therefore not been established.

Available covariance. The analysis does not include full covariance for the tomography models or the complete GRACE coefficient covariance. The bootstrap measures how stable the result is when different models are selected from the chosen families. It does not represent the full uncertainty in Earth's density field.

4.2 What does not change the primary result

Reversing the global sign does not move the principal axes. It only switches the minimum and maximum roles. Under both the adopted and reversed sign cases, N_p' remains outside the pooled bootstrap region. The primary classification therefore stays `unsupported`.

Choosing the candidate-facing end of an axis for a figure also does not change the result, because all distances are calculated as axial distances. Close results from individual tomography models, static gravity, or GRACE remain secondary comparisons. They do not replace the pooled containment result.

4.3 Pass, fail, and indeterminate conditions

The final decision rule is:
Pass. The candidate lies inside the declared bootstrap region around the family-balanced lowermost-mantle center.
Fail. The candidate lies outside the empirical 99% bootstrap region.
Indeterminate. The required coordinates, source information, axis-role mapping, tensor data, or uncertainty results are missing.

The candidate meets the fail condition for this evidence class.

Table 6: GRACE and GRACE-FO mission-scale geometry. Intervals come from 1,000 block bootstrap realizations.

Representation	Median difference	Approximate 95% interval
Named minimum role	88.83°	86.01-89.95°
Named intermediate role	4.59°	0.82-11.60°
Named maximum role	88.78°	85.97-89.95°
Optimally permuted unordered frame	4.78°	1.45-11.83°

4.4 What the analysis cannot establish

This analysis does not show:

- that the deep mantle, lithosphere, static gravity field, and GRACE are physically coupled;
- that one principal axis continues through different depths or across time;
- the path of a pole shift;
- whether enough torque, energy, or angular impulse is available for the proposed motion;
- that the candidate coordinate was occupied in the past;
- whether the candidate agrees with paleomagnetic, geological, climate, archaeological, or plate-kinematic evidence.

Those are separate evidence classes or dynamics questions.

5 Conclusions

This study calculated and compared preferred-axis geometry from mantle mass anomaly models, static gravity, the solid lithosphere, and time variable gravity measured by GRACE and GRACE-FO. The primary test estimated a family balanced preferred-axis center for the lowermost mantle and compared it with the proposed pole. The analysis includes three lowermost mantle null tests with 10,000 realizations each, 2,000 family bootstrap samples, 1,000 GRACE and GRACE-FO bootstrap samples, cross-validation, sign and axis-role sensitivity tests, static gravity residual comparisons, a solid lithosphere comparison, a frozen candidate registry, axial plotting rules, and validation checks.

The results show a statistically unusual preferred-axis cluster in the lowermost mantle. The family balanced center is near $1.02°\text{S}, 17.24°\text{E}$ in the candidate facing representation. This center remains fairly stable when different models are selected from the available model families. The bootstrap uncertainty is based on a small set of discrete model combinations and does not represent the full physical uncertainty in Earth's mantle structure.

The pooled lowermost mantle result does not recover N_p'. The candidate lies 18.81° from the family-balanced center and outside the empirical 99% bootstrap radius. Some individual comparisons lie closer. The primary static gravity maximum axis is 9.45° away, and one UNICA25 lowermost-mantle tensor is 2.16° away.

This conclusion could change if new independent tomography models produce a different family balanced center, a more complete uncertainty model expands the bootstrap region enough to include the candidate, a better constrained static gravity residual places the physical maximum-moment axis near the candidate, or another gravity or inertia dataset independently recovers the same location. A reliably tracked GRACE branch would also improve the time variable comparison. Physical coupling would still need to be tested separately before treating the different layers as one connected system.

Data and code availability

The public-facing report is intended to provide the datasets used, assumptions, equations, tensor setup, null constructions, validation logic, final tables, figures, interpretation, and limitations. Working notebooks, cleaned intermediate datasets, helper functions, and the full preprocessing pipeline are retained as private research materials. Machine readable result tables may be distributed with the report where appropriate.

References

[1] R. B. Cunningham, *Inversion: ECDO Theory—The Hidden Mechanism Driving Cataclysm, Cultural Tradition, and Climate*. Atlanta, GA: Meridian Standard Press, 2026.

[2] The Ethical Skeptic, "On Modeling ECDO Theory," 20 June 2026. Available: https://theethicalskeptic.com/2026/06/20/on-modeling-ecdo-theory/.

[3] A. M. Dziewonski and D. L. Anderson, "Preliminary reference Earth model," *Physics of the Earth and Planetary Interiors*, vol. 25, pp. 297–356, 1981. doi: 10.1016/0031-9201(81)90046-7.

[4] M. E. Pasyanos, T. G. Masters, G. Laske, and Z. Ma, "LITHO1.0: An updated crust and lithospheric model of the Earth," *Journal of Geophysical Research: Solid Earth*, vol. 119, pp. 2153–2173, 2014. doi: 10.1002/2013JB010626.

[5] J. Ritsema, A. Deuss, H. J. van Heijst, and J. H. Woodhouse, "S40RTS: A degree-40 shear-velocity model for the mantle from new Rayleigh-wave dispersion, teleseismic traveltime, and normal-mode

splitting-function measurements," *Geophysical Journal International*, vol. 184, pp. 1223–1236, 2011. doi: 10.1111/j.1365-246X.2010.04884.x.

[6] S. P. Grand, "Mantle shear-wave tomography and the fate of subducted slabs," *Philosophical Transactions of the Royal Society A*, vol. 360, no. 1800, pp. 2475–2491, 2002. doi: 10.1098/rsta.2002.1077.

[7] IRIS Data Management Center, "TX2011: A global mantle shear-wave tomography model," Earth Model Collaboration data product, 2011. doi: 10.17611/DP/9991721.

[8] S. Durand, E. Debayle, Y. Ricard, C. Zaroli, and S. Lambotte, "Confirmation of a change in the global shear-velocity pattern at around 1000 km depth," *Geophysical Journal International*, vol. 211, no. 3, pp. 1628–1639, 2017. doi: 10.1093/gji/ggx405.

[9] A. Tesoniero, L. Auer, L. Boschi, and F. Cammarano, "Hydration of marginal basins and compositional variations within the continental lithospheric mantle inferred from a new global model of shear and compressional velocity," *Journal of Geophysical Research: Solid Earth*, vol. 120, 2015. doi: 10.1002/2015JB012026.

[10] IRIS Data Management Center, "SPani: 3-D joint anisotropic compressional- and shear-wave velocity model of the whole mantle," Earth Model Collaboration data product, 2015. doi: 10.17611/DP/EMC-SPANI.

[11] S. W. French and B. Romanowicz, "Whole-mantle radially anisotropic shear-velocity structure from spectral-element waveform tomography," *Geophysical Journal International*, vol. 199, no. 3, pp. 1303–1327, 2014. doi: 10.1093/gji/ggu334.

[12] S. Noe et al., "The Collaborative Seismic Earth Model: Generation 2," *Journal of Geophysical Research: Solid Earth*, 2024. doi: 10.1029/2024JB029656.

[13] Collaborative Seismic Earth Model developers, "CSEM2-Globe VSV and RHO model products," Generation-2 seismic-velocity and density-related fields, 2024. The exact frozen files and checksums used in this study are recorded in the project provenance registry.

[14] UNICA25 model developers, "UNICA25 absolute P-wave mantle model product," frozen absolute-V_p volume and derived equal-area depth-anomaly product, 2026. The exact source file, version, and checksum used in this study are recorded in the project provenance registry.

[15] GOCO Consortium, "GOCO2025s global static gravity-field model," fully normalized spherical-harmonic coefficient product, International Centre for Global Earth Models, 2025. Available: https://icgem.gfz-potsdam.de/.

[16] G. Petit and B. Luzum, eds., *IERS Conventions (2010)*. IERS Technical Note 36, International Earth Rotation and Reference Systems Service, Frankfurt am Main, Germany, 2010. Available: https://iers-conventions.obspm.fr/.

[17] B. D. Tapley, S. Bettadpur, M. Watkins, and C. Reigber, "The Gravity Recovery and Climate Experiment: Mission overview and early results," *Geophysical Research Letters*, vol. 31, L09607, 2004. doi: 10.1029/2004GL019920.

[18] F. W. Landerer et al., "Extending the global mass change data record: GRACE Follow-On instrument and science data performance," *Geophysical Research Letters*, vol. 47, e2020GL088306, 2020. doi: 10.1029/2020GL088306.

[19] GFZ German Research Centre for Geosciences, "GRACE Level-2 Monthly Gravity Field Solutions, GFZ Release 06 (GSM Product)," GFZ Data Services. doi: 10.5880/GFZ.GRACE_06_GSM.

[20] GFZ German Research Centre for Geosciences, "GRACE Follow-On Level-2 Monthly Gravity Field Solutions, GFZ Release 06.3 (GSM Product)," GFZ Data Services. doi: 10.5880/GFZ.GRACEFO_06_GSM.

[21] GFZ German Research Centre for Geosciences, "GRAVIS RL06 Satellite-Laser-Ranging Replacement Series for $\bar{C}_{20}$," GFZ Data Services. doi: 10.5880/GFZ.GRAVIS_06_C20_SLR.

[22] D.-N. Yuan, *GRACE Follow-On Level-2 Gravity Field Product User Handbook*. Jet Propulsion Laboratory, JPL D-103922, 2019.

[23] GOCO Consortium, "GOCO2025s formal spherical-harmonic coefficient standard deviations," distributed with the GOCO2025s `.gfc` coefficient product, International Centre for Global Earth Models, 2025. The present analysis used the reported diagonal degree-2 standard deviations; unavailable coefficient correlations and uncertainty in the applied correction vectors were not included.

[24] S. Buchanan, "Does Gravity See Slow Mantle Regions as Heavy or Light? A Global-Scale Comparison of Mantle Models with Observed Gravity," independent geophysical modeling case study, July 2026. Available: https://github.com/stephanieskystory/slow-mantle-gravity-sign-test/tree/main/paper.